

Interacting Multipoles of Rank 2 and 4:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}\cdot$$

$$\begin{aligned}
& +105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\gamma,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\epsilon)\cdot\delta(\gamma,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\beta)\cdot\delta(\epsilon,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\epsilon)\cdot\delta(\beta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\zeta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\delta)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta} \\
& +105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\epsilon)\cdot\Theta_{\alpha\beta}\cdot\Phi_{\gamma\delta\epsilon\zeta}
\end{aligned}$$

$$(1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}.$$

[illegible]

$$\begin{aligned} & \xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\gamma \cdot R_\gamma \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\delta\delta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\epsilon\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\delta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\delta\delta\zeta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\beta \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\gamma\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\epsilon\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\delta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\delta\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\epsilon} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\delta\delta\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\gamma \cdot R_\zeta \cdot \Theta_{\alpha\beta} \cdot \Phi_{\alpha\beta\gamma\zeta} \\ & +105 \cdot R_z^4 \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\epsilon} \end{aligned} \quad (2b)$$

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta}=R_z^{-13}\cdot\frac{1}{315}.$$

[illegible]

Simplify Quadrupoles:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{array}{l} +10395 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} - 945 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1890 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} - 2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -3780 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} - 3780 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} \\ -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zzzz} + 105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} + 210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ +420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} + 630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} + 420 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} \\ +630 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} + 105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} - 15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} - 90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{array} \right] \quad (4a)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{array}{l} +2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} - 525 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\epsilon\epsilon} \\ -1050 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\delta\delta} - 1575 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zz\gamma\gamma} \\ -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zzzz} + 105 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ +210 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{\gamma\gamma\delta\delta} + 105 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} + 210 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\delta\delta} \\ +315 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\gamma\gamma} - 15 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\epsilon\epsilon} \\ -30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} - 30 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\epsilon\epsilon} \\ -60 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\delta\delta} - 90 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{\alpha\alpha\gamma\gamma} \end{array} \right] \quad (4b)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \left[+2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha} \cdot \Phi_{zz\alpha\alpha} \right] \quad (5)$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = R_z^{-13} \cdot \frac{1}{315} \cdot \left[+2835 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} + 1260 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxxz} \right] \quad (6a)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[+4095 \cdot R_z^6 \cdot \Theta_{zz} \cdot \Phi_{zzzz} + 1260 \cdot R_z^6 \cdot \Theta_{xx} \cdot \Phi_{xxxz} \right. \\ \left. +1260 \cdot R_z^6 \cdot \Theta_{yy} \cdot \Phi_{yyzz} \right] \quad (6b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Theta_{\alpha\beta}\Phi_{\gamma\delta\epsilon\zeta} = \left[\begin{array}{l} +13 \cdot R_z^{-7} \cdot \Theta_{zz} \cdot \Phi_{zzzz} + 4 \cdot R_z^{-7} \cdot \Theta_{xx} \cdot \Phi_{xxxz} \\ +4 \cdot R_z^{-7} \cdot \Theta_{yy} \cdot \Phi_{yyzz} \end{array} \right] \quad (7)$$